Celal Tarık Bayraktar

Software Developer · C#/ASP.NET Core · React · SQL

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SUMMARY

Software developer and electrical-electronics engineer, building full-stack web applications in C#/ASP.NET Core, React and SQL at Türkiye's Ministry of Trade, and supporting its customs transit (NCTS) data operations in Oracle. Outside work, ships products end to end by directing Claude Code: a web-based SPICE circuit simulator, two live iPhone games and a Unity simulation in development. Data Science MSc student at Gazi University.

EXPERIENCE

Software Developer (Assistant Trade Expert) Aug 2025 – present

Ministry of Trade, General Directorate of Information Technologies (BTGM) · Ankara

- Modernize the ministry's ETAP and TEKİZ web platforms with React, C#/ASP.NET Core Web API and SQL.
- Build React screens and components, and the ASP.NET Core services, controllers and business logic behind them.
- Design new SQL tables and reshape existing schemas for the redesigned workflows.
- Diagnose and fix production defects, and improve existing features.
- Support customs transit operations under NCTS: write Oracle queries in Toad to trace status and data-consistency problems, and resolve records that stall in the expected flow, working with the General Directorate of Customs.

Engineering Intern Aug – Sep 2022

Koçak Mühendislik · Aksaray

- Supported the engineering team with calculations, technical drawings and site visits.

PROJECTS

PlainSpice · Circuit simulator for the web · plainspice.com 2026

- Beginner-first circuit simulator for students and my Solved Circuits channel: ngspice 47 compiled to WebAssembly, matching desktop ngspice to six significant figures, with the results written on the drawing.
- React 19 and TypeScript: schematic editor synced with the netlist, plain-language checks for beginner mistakes, share links, offline use; WCAG 2.2 AA, Playwright and axe tests, CI to GitHub Pages.

One More Passenger · iPhone game · live on the App Store 2026

- Physics stacking game: a crane swings passengers over a rocking ferry on a voyage from Istanbul to Bodrum, and every landing shifts the balance; a daily challenge is the same for every player.
- HTML5 canvas game in an Expo (React Native) app, built in the cloud without a Mac; English and Turkish; AdMob ads with Google's consent flow; 254 end-to-end tests and bot playtests for balance.

Don't Spill: Coffee Delivery · iPhone game · live on the App Store 2026

- One-touch physics game built with web technologies and Capacitor, with iOS builds from Windows through a cloud build service.
- Launched in October 2026 and shipped v1.1 within two weeks; in-app purchases (tips and costume sets).

From Flip-Flops to Atari · Unity (C#) simulation game In development

- Educational game: build a computer from logic gates up to a console, on a deterministic C# engine.

Educational YouTube channels · Automated production pipelines 2026

- Two channels (computer internals, worked circuit-analysis problems) whose scripting, narration, rendering and publishing run as scheduled jobs.

SKILLS

Languages	C#, JavaScript, Python, SQL, C++, MATLAB
Backend	ASP.NET Core Web API, .NET, REST APIs
Frontend and mobile	React, HTML/CSS, Capacitor, Expo, iOS App Store publishing (TestFlight, in-app purchases)
Data	Oracle (Toad), SQL schema design, pandas, scikit-learn, spaCy, NLTK
Game development	Unity (C#)
Tools	Git/GitHub, Visual Studio, VS Code, AI-assisted development (Claude Code)

EDUCATION

MSc, Data Science · Gazi University Feb 2026 – present

BSc, Electrical-Electronics Engineering · Ankara University 2023

LANGUAGES

English Advanced (YDS 93.75/100, 2024)

Turkish Native